MILITARY SPECIFICATION

FIRING DEVICE, EXPLOSIVE ACTUATOR

(CCU-74/B)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the CCU-74/B Explosive Actuator Firing Device, referred to herein as the device.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-B-85251 Bomb, Binary Chemical, Empty (BLU-80/B)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1377

AMSC N/A

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-453	Inspection, Radiographic.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

OTHER GOVERNMENT DOCUMENTS

Code of Federal Regulations (CFR)

49 CFR 171-178	Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402.)

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1433AS864	Firing Device, Explosive Actuator, CCB-74/B.

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SA1433AS1151	Firing Mechanism Fixture Firing Device.
SA1433AS1152	Firing Device Fixture Test Vibration.
SA1433AS1166	Dent Block.

(Copies of specifications, standards, handbooks, drawings, publications and Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Item description. The device consists of a housing and a lanyard-activated firing pin.

3.2 Characteristics.

3.2.1 Performance. During tail fin deployment, when a load is applied to the firing device lanyard such that the firing pin retracts and releases, the firing pin shall obtain a kinetic energy of 45 to 125 inch-ounces when the firing pin reaches the end of the firing device housing.

3.2.2 Environmental. The device shall not be damaged or its performance degraded after exposure to the environments specified in 3.2.2.1 through 3.2.2.5.

3.2.2.1 High temperature. The device shall withstand exposure to a high temperature of +74 degrees Celsius (°C) for 48 hours.

3.2.2.2 Low temperature. The device shall withstand exposure to a low temperature of -48°C for 24 hours.

3.2.2.3 Temperature shock. The device shall withstand alternate sequential exposure to temperatures of -48°C and +74°C.

3.2.2.4 Mechanical shock. The device shall withstand exposure to half sine wave impact shocks having a peak amplitude of 15 gravity units (g) and a total duration of 11 milliseconds (ms).

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3.2.2.5 Vibration. The device shall withstand exposure to the vibration spectrum specified on Figure 1 in each of three mutually perpendicular axes.

3.3 Design and construction.

3.3.1 Production drawings. The device shall be fabricated and assembled in accordance with Drawing 1433AS864 and associated documents, and as specified herein.

3.3.1.1 Radiographic. The device shall show no evidence of defects and the internal construction shall be in accordance with Drawing 1433AS864 and associated documents.

3.3.2 Standards of manufacture.

3.3.2.1 Certification. When specified in the contract or purchase order (see 6.2.2), the contractor shall prepare certification for the procuring activity, accompanied by objective quality evidence in accordance with MIL-STD-109, that the materials, processes, and parts used in the assembly meet the applicable requirements for each item.

3.3.2.2 Working environments. Any special working environments for the manufacture of the device shall be in accordance with MIL-B-85251.

3.3.3 Marking. Each device shall be marked in accordance with MIL-STD-130, with the manufacturer's name or symbol, part number, and manufacturer's lot designation. The markings shall be in accordance with Drawing 1433AS864 and associated documents.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.2.3).

3.5 Workmanship. The device shall be fabricated in a manner that ensures compliance with all requirements of this specification and Drawing 1433AS864. Unless otherwise specified herein or on Drawing 1433AS864, fabrication and assembly practices shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

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unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

4.2 First article. When specified in the contract or purchase order (see 6.2.3), a first article inspection shall be performed on 15 devices manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the first article sample shall be at the risk of the contractor.

4.2.1 First article inspections. First article inspections shall be conducted on the first article sample and shall consist of the inspections of Table I in the sequence shown.

4.2.1.1 Acceptance criteria. The failure of one or more devices to pass any of the first article tests and inspections of Table I shall cause rejection of the first article sample.

4.2.1.2 First article inspection report. When specified in the contract or purchase order (see 6.2.2), a first article inspection report shall be prepared for the procuring activity. The report shall contain the following information accompanied by certification (see 6.2.2) which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the first article sample complies with all quality assurance provisions of this specification for the device specified in the contract or purchase order.

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b. Number of units of product inspected.

c. Results obtained for all inspections performed.

d. Purchase descriptions or contract number and date together with an identification and date of changes.

e. Certificates of compliance of all material procured directly by the contractor.

f. Date submitted.

The certification shall be signed by a responsible agent of the certifying organization. The initial certification shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certifications unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

TABLE I. First article inspections.

Inspections	Requirement paragraph	Method paragraph
Group I (100 percent)		
Visual/radiographic examination	3.3.1, 3.3.1.1, 3.5	4.5.1, 4.5.2
Group II		
High temperature	3.2.2.1	4.6.1
Low temperature	3.2.2.2	4.6.2
Temperature shock	3.2.2.3	4.6.3
Mechanical shock	3.2.2.4	4.6.4
Vibration	3.2.2.5	4.6.5
Performance	3.2.1	4.5.3

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4.2.1.3 First article verification. At the discretion of the procuring activity, first article tests, or any portion thereof, shall be repeated under any of the following conditions:

a. The manufacture has modified the product (such as change of raw materials, processes, production procedures, or methods).

b. Where there is evidence that the quality of the product has not been maintained. This evidence may be in the form of accumulated product failures, of systems failures attributable to the product, or failure of the product to pass any of the tests for production lot acceptance that may be conducted by or for the procuring activity.

c. When applicable documents have been amended or revised sufficiently so that continued validity of the previous first article testing is questionable.

d. When the manufacturer has not produced the product to this specification for a period of 12 months or longer.

e. When the manufacturer changes the location at which the product is produced.

4.3 Quality conformance inspections. The quality conformance inspections shall be performed on each inspection lot and shall consist of the examination and tests listed in Table II in the sequence shown.

TABLE II. Quality conformance inspections.

Inspections	Requirement paragraph	Method paragraph
Group I (100 percent)		
Visual/radiographic inspection	3.3.1, 3.3.1.1, 3.5	4.5.1, 4.5.2
Group II (sample)		
Vibration	3.2.2.5	4.6.5
Performance	3.2.1	4.5.3

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4.3.1 Inspection lot. An inspection lot shall be in accordance with MIL-STD-105. An inspection lot shall be formed from devices which have passed Group I inspections.

4.3.2 Sampling. A sample shall be randomly selected from each inspection lot. The sample size shall be in accordance with MIL-STD-105 Normal Inspection, Inspection Level II, Single Sampling, and an acceptable quality level of 0.65 percent defective.

4.3.3 Acceptance criteria. Failure of one or more devices in the sample to pass any of the Group II quality conformance tests shall cause rejection of the inspection lot represented.

4.3.4 Quality conformance inspection report. When specified in the contract or purchase order (see 6.2.2), a quality conformance inspection report shall be included with each lot. Reports shall contain the following information accompanied by certification (see 6.2.2) which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of this specification for the device specified in the contract or purchase order.

b. Number of units of product inspected.

c. Results obtained for all inspections performed.

d. Contract or purchase order number and date together with an identification and date of changes.

e. Certificates of compliance of all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certification shall be signed by a responsible agent of the certifying organization. The initial certification shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certification unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.4 Test equipment and conditions.

4.4.1 Standard test equipment. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements under the conditions and to the accuracies specified herein. All measuring and test equipment used by the contractor for the performance of the inspections specified herein shall be calibrated and maintained in accordance with MIL-STD-45662. Test equipment, including associated calibration programs and operation and maintenance procedures shall be subject to approval by the procuring activity. When special test equipment and circuits are devised or commercially available equipment is employed, all test equipment circuits and methods shall be subject to approval by the procuring activity. Unless otherwise specified herein, the magnitude of any error introduced by test equipment shall be not greater than 10 percent of the tolerance of the requirement being measured.

4.4.2 Test conditions. Unless otherwise specified herein, all tests shall be conducted at ambient temperature and relative humidity. Where special conditions of tests are specified, equipment employed shall be capable of providing those specified conditions and all monitoring and measurement devices shall meet the requirements of 4.4.1. When special conditions of tests are not specified, commercially available direct measurement equipment may be employed provided they meet the requirements of 4.4.1.

4.5 Inspection methods.

4.5.1 Visual inspection. The device shall be visually inspected to determine conformance to Drawing 1433AS864 and associated documents. Any device not meeting the requirements of 3.3.1 shall be rejected.

4.5.2 Radiographic inspection. Radiographic inspection of the device shall be in accordance with MIL-STD-453. Any device not meeting the requirements of 3.3.1.1 shall be rejected.

4.5.3 Performance. The energy requirements of 3.2.1 shall be measured using the test fixture shown on Drawing SA1433AS1151 (see Figure 2). Failure of one or more of the devices in the sample to meet the requirements of 3.2.1 shall cause rejection of the lot represented.

4.6 Environmental.

4.6.1 High temperature. The device shall be subjected to the high temperature test of MIL-STD-810, Method 501.1, Procedure I (see 6.3), except the temperature shall be +74°C. Steps 4 and 5 shall be omitted.

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4.6.2 Low temperature. The device shall be subjected to the low temperature test of MIL-STD-810, Method 502.1, Procedure I (see 6.3), except the temperature shall be -48°C. Steps 4 and 5 shall be omitted.

4.6.3 Temperature shock. The device samples shall be subjected to the temperature shock test of MIL-STD-810, Method 503.1, Procedure I (see 6.3), except the high temperature shall be +74°C and the low temperature shall be -48°C.

4.6.4 Mechanical shock. Testing shall be performed in accordance with MIL-STD-810, Method 516.2, Procedure I (see 6.3), with the following modifications. Testing shall be conducted by applying three shocks in each direction through three mutually perpendicular principle axes.

a. The shock pulse signature shall be a half sine pulse, 15 g peak and 11 ms duration. The shock pulse tolerance limits shall be in accordance with MIL-STD-810, Method 516.2, Figure 516.2-2 (see 6.3).

b. Before applying mechanical shocks, the device shall be temperature conditioned at -48 ±3°C for not less than 4 hours. If the device is removed from the chamber to apply the required shocks, the elapsed time between removal from the chamber and the start of application of any shock along any axis may be increased to 5 minutes, provided the device is not exposed to ambient temperature for more than 3 minutes of the 5-minute period. If shock testing is not completed within the time limits specified, the device shall be returned to the conditioning chamber for not less than 1 hour before shock tests are resumed.

4.6.5 Vibration. Testing shall be conducted in accordance with MIL-STD-810, Method 514.2 (see 6.3). Testing shall be conducted at a temperature of -48 ±3°C for 30 minutes per axis. The vibration test profile shall be in accordance with Figure 1. The tolerance on vibration test levels shall be in accordance with MIL-STD-810, Method 514.2, Random Vibration Tests (see 6.3).

4.7 Packaging, packing, and marking. Prior to shipment, examination shall be made to assure that packaging, packing, and marking conform to Section 5 of this specification.

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5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be Level C in accordance with FED-STD-102. The device shall be packaged in a manner to ensure delivery to the first destination without damage to the device and shall be in accordance with 49 CFR 171-178.

5.2 Packing. Packing shall be Level C in accordance with FED-STD-102. The device shall be packed in containers in accordance with 49 CFR 171-178.

5.3 Marking. In addition to any required special markings required by the contract or purchase order (see 6.2.1), shipping containers and palletized loads shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The device is intended to open the tail fins of a binary chemical bomb.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspection and inspection facilities if different than 4.1.

c. Special markings, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the contractor in accordance with the contract or purchase order requirements. Deliverable data requierd by this specification are cited in the following paragraphs.

Paragraph	Data requirement	Applicable DID	Option
3.3.2.1, 4.2.1.2, 4.3.4	Certificate of Compliance	DI-E-2121	---

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Paragraph	Data requirement	Applicable DID	Option
4.1.2.2	First Article Inspection Report	DI-T-4902	---
4.3.4	Inspection Report	DI-T-2072	---

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the item should a first article sample. The first article should consist of 15 devices (see 4.2). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who whish to rely on such production or test, should furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Environmental testing. MIL-STD-810C should be used when performing the environmental tests of 4.6.1, 4.6.2, 4.6.3, 4.6.4, and 4.6.5.

Preparing activity:
Navy - AS

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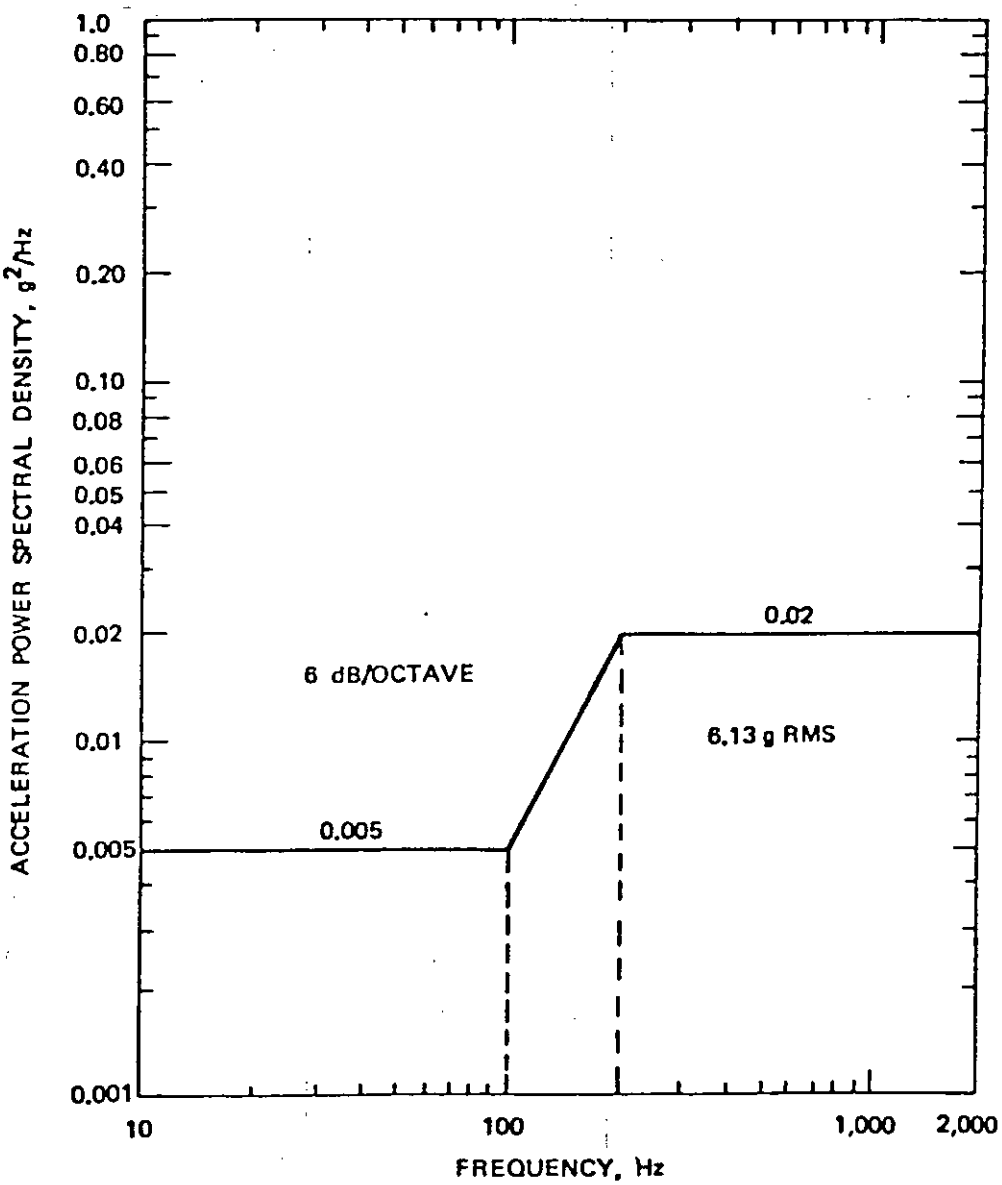


FIGURE 1. Vibration spectrum.

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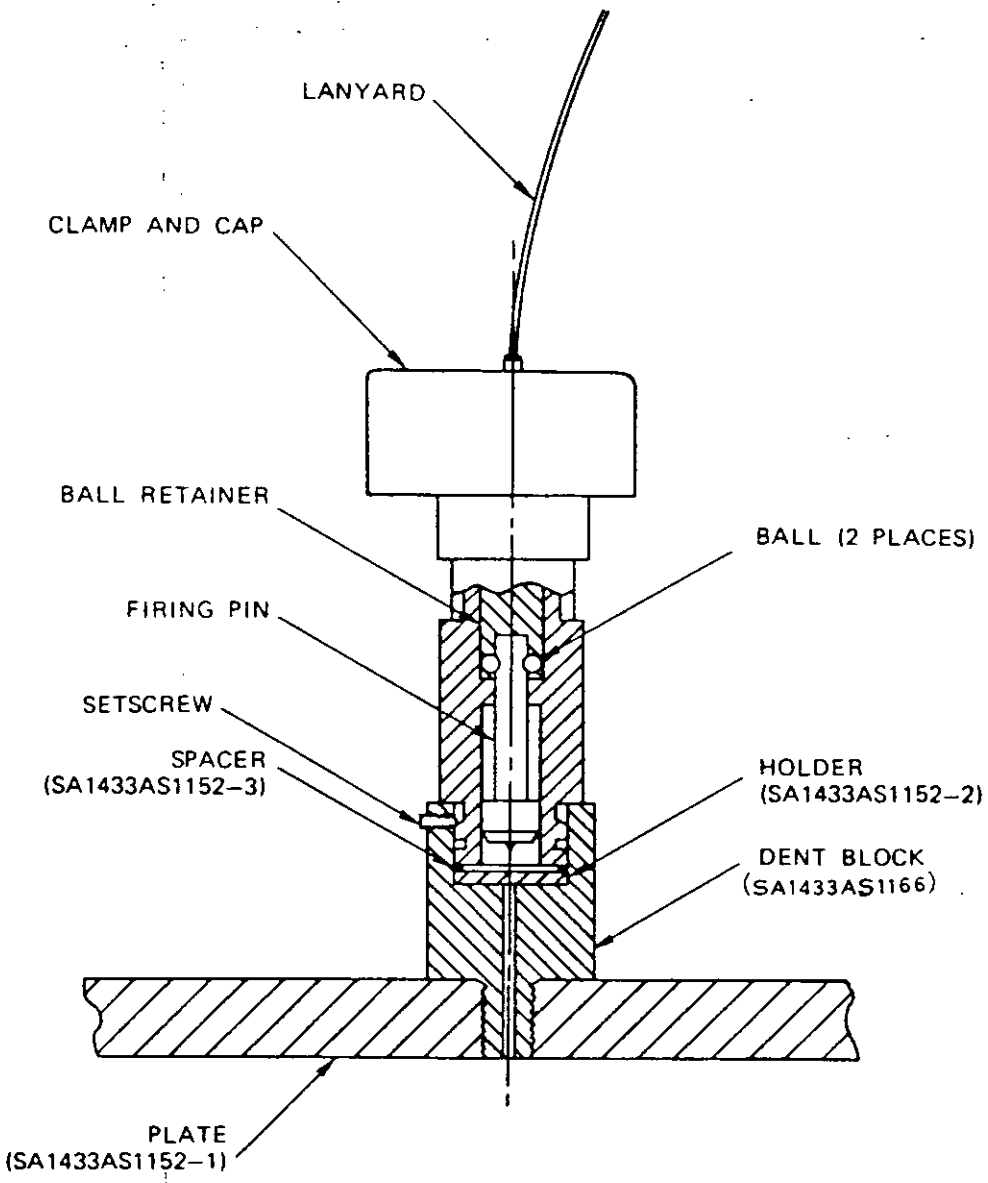


FIGURE 2. Test set up.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-85531(AS)	FIRING DEVICE, EXPLOSIVE ACTUATOR (CCU-74/B)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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